LSST DP1 Catalogue Columns and Flag Filters

This document provides a brief outline of the necessary columns to extract from LSST's DP1 to create the smaller dataset used for cross-matching. It also provides more information on the "flag" index, additional columns to extract and combine into a quality flag value.

Each catalogue will need to be transformed from its current state in the DAC to separate files for ingestion into the cross-match process, to allow for a common interface for flexible matching across multiple different catalogues. A summary of this in its current form is provided below.

NOTE: we currently assume that LSST is filtered for primary detections, but this should be checked regularly to ensure that no filtering on secondary detections is needed.

ID

The unique designation `objectID` .

Sky coordinates, uncertainty

The sky positions are `coord_ra` and `coord_dec` respectively.

To create a single astrometric uncertainty, we convert the covariance matrix to semi-major and semi-minor axes and take an average:

```
sig_maj = sqrt(0.5 * (coord_raErr**2 + coord_decErr**2 +
                      sqrt((coord_raErr**2 - coord_decErr**2)**2 +
                           4 * coord_ra_dec_Cov**2)))
```

and

```
sig_min = sqrt(0.5 * (coord_raErr**2 + coord_decErr**2 -
                      sqrt((coord_raErr**2 - coord_decErr**2)**2 +
                           4 * coord_ra_dec_Cov**2))).
```

We wish to take the geometric mean (instead of the arithmetic mean),

$$\left(\prod_{i=1}^{n} \sigma_i\right)^{\left(\frac{1}{n}\right)},$$

to preserve the ellipse area, so

```
sig_avg = sqrt(sig_maj * sig_min).
```

NOTE: all uncertainties are provided in degrees, but we wish for them to be in arcseconds, so we must perform a final unit conversion of `sig_avg = sig_avg * 3600` .

Magnitudes

We require magnitudes for all six filters. However, here we wish to use two separate algorithms for flux (and thus magnitude) determination, suited to different science cases: the `cModel` magnitudes (the best option for non-crowded galaxy fluxes) and `psf` magnitudes (the best option for crowded stellar fields).

We extract `*_QFlux` , where `*` is any one of the filters (`ugrizy`) and `Q` is either `cModel` or `psf` , giving six times two *fluxes*. We additionally need `*_QFlux_flag` , for a total of two columns per filter per algorithm.

We use `psf` within a Galactic latitude of 20 degrees ($|b| \leq 20$), and outside of the Galactic plane we use `cModel` , for all six filters.

NOTE: The LSST fluxes are provided in nanoJanskys, but we wish to use magnitudes. The conversion from nJy to AB magnitudes should be straightforward enough, however, using:

```
import astropy.units as u
mag = (flux * u.nJy).to(u.ABmag)
```

Signal-to-noise ratios

SNRs can be calculated following a similar pattern to the magnitudes. We extract `*_QFlux` and `*_QFluxErr` – where `*` will be any one of the `ugrizy` bands, and `Q` is either `psf` or `cModel` – and take the ratio of flux to fluxerr.

Best detection index

The following quality cuts apply to the photometric bands, with a default flag number of `2` unless criteria are met:

Photometric Flag	Flag Number	Criteria
Non-detection	-1	`*_magerr`[1] == NaN OR `*_magerr` <= 0 OR `*_flag` is `True` OR `*_pixelflags_edge` is `True` OR `*_pixelflags_offimage` is `True`
Low quality (saturated)	0	`*_pixelflags_saturated` is `True`
Low quality (generic)	1	`*_magerr` >= 0.5 OR `*_pixelflags_bad` is `True` OR `*_pixelflags_clipped` is `True` OR `*_pixelflags_cr` is `True` OR `*_pixelflags_sensor_edge` is `True` OR `*_pixelflags_suspect` is `True`

Any source with entirely negative flags is removed; if no filter is at least a zero-flagged detection then we assume the object should be discarded. For best detection filter, we prioritise filters in order of `irzgyu`. First we check for `2`-flagged filters, then `1`-flagged filters and finally `0`-flagged photometry. It would be most sensible to define the order of the filters in increasing wavelength order, and hence have the columns in order of `ugrizy`, with `u` in `array[0]` (or `array[1]` for one-indexing) and `y` in `array[5]` (`array[6]`).

Epoch and proper motion

Each filter has its own epoch column, `*_epoch`. Record the epoch of the filter that was selected as the "best detection" filter for each object. LSST's DP1 has no proper motion information, so otherwise save no information beyond the epoch.

Zone overlap index

A flag for "is this the primary zone this object is in, or is this object part of the padding around this zone", required for all cases where chunking is performed to parallelise the cross-matching process.

Reject catalogue

Any object that is removed from the final catalogue should be placed in a separate "reject" catalogue and provided separately. As these will fail to meet the best detection criterion that does not need storing; however, we still will require the zone overlap index.

[1] The use of `*_mag` and `*_magerr` refers to the selection of either `cModel` or `psf` as `Q` in `*_QFlux` and the conversion from flux to magnitude; `*_flag` is shorthand for `*_QFlux_flag`, again choosing the appropriate `Q` for the galactic coordinates, and so on for the `pixelflags` parameters.